



Laboratorio No.1 – Semana 14 - Integración servicios en la nube

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Objetivo: Desarrollar y publicar una aplicación web que permita interactuar con archivos almacenados en Azure Blob Storage, utilizando control de versiones con Git y una pipeline (script) CI/CD de Azure DevOps para automatizar el despliegue de la aplicación en Azure App Service.

Resultado esperado

- Aplicación web funcional ejecutándose en Azure App Service.
- Conexión entre App Service y Azure Blob Storage.
- Listado y consulta de archivos almacenados en un contenedor Blob.
- Operación de carga de archivos al Blob Storage.
- Repositorio Git con historial de cambios.
- Pipeline de Azure DevOps que compile y despliegue automáticamente la aplicación.
- Evidencias que demuestren el funcionamiento de la solución.

Repositorio recomendado

Utilice como punto de partida el ejemplo oficial de Microsoft/Azure Samples revisado para esta actividad:

GitHub: <https://github.com/Azure-Samples/serviceconnector-webapp-storageblob-dotnet>

El tutorial oficial utiliza una aplicación web .NET conectada a Azure Blob Storage mediante una identidad administrada y Service Connector. La documentación indica además que la aplicación puede desplegarse en Azure App Service.

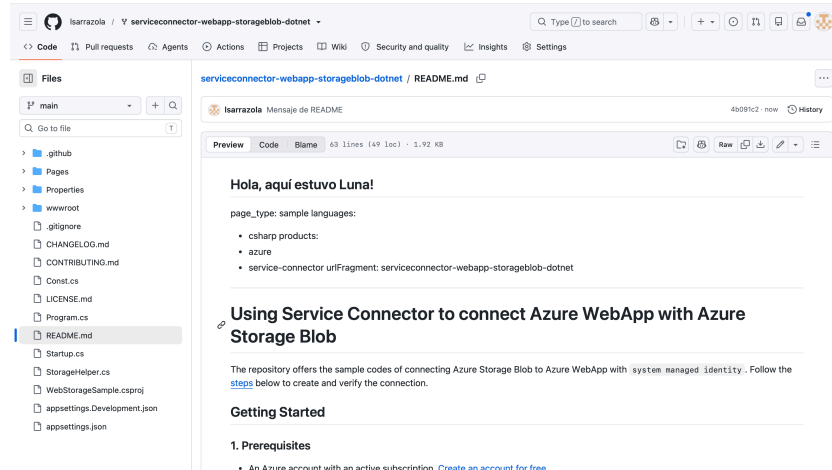
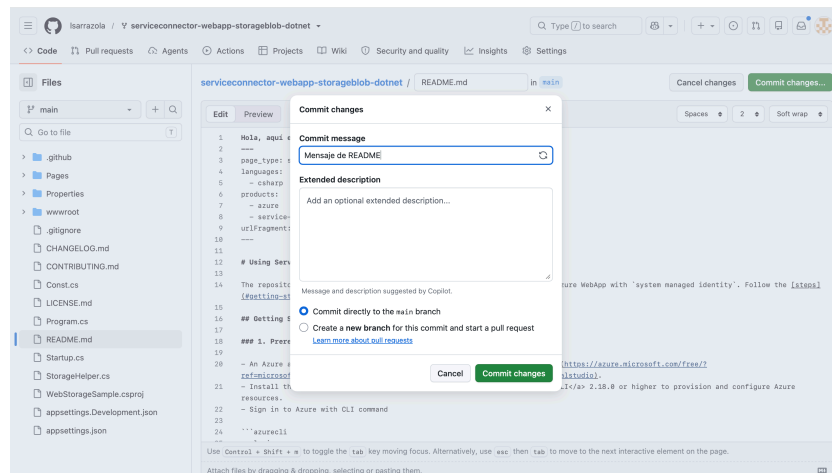
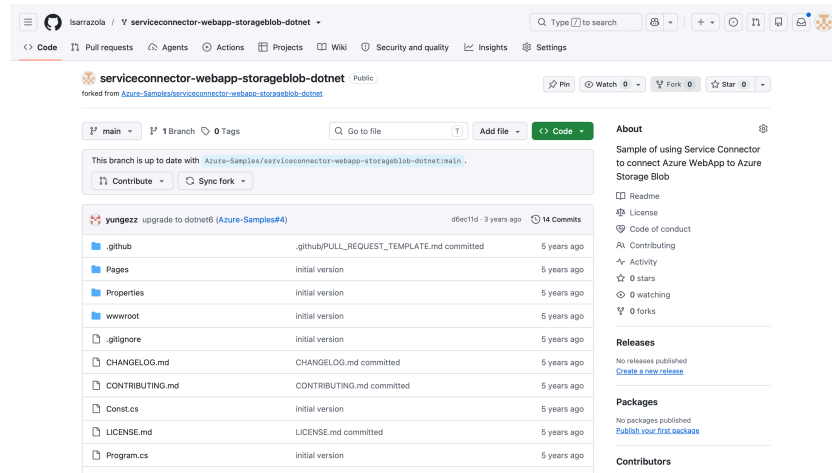
Tecnologías

Tecnología	Uso
C# / .NET	Desarrollo de la aplicación web.
Azure Blob Storage	Almacenamiento de archivos.
Azure App Service	Hospedaje de la aplicación.
Git	Control de versiones.
Azure DevOps Pipelines	Integración y despliegue continuo.
Managed Identity / Service Connector	Acceso seguro desde App Service al almacenamiento.

Realización de la actividad

1. Preparar el repositorio

En el repositorio brindado arriba, realice un fork del repositorio base o copie el proyecto a un repositorio propio. El repositorio debe pertenecer al equipo de trabajo. Realice un primer commit identificable.



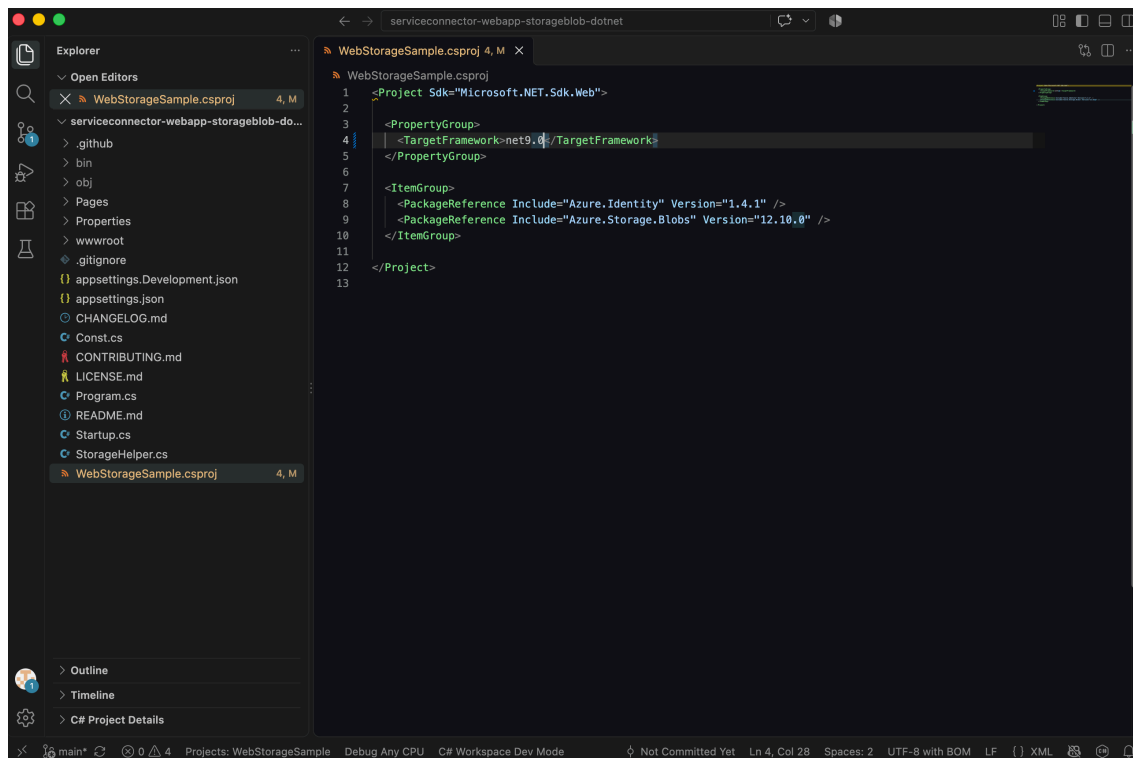
2. Revisar la aplicación:

Ejecute la aplicación localmente y revise cómo obtiene acceso al Blob Storage. Identifique el punto del código donde se realiza la lectura/escritura de blobs y documente brevemente su funcionamiento. Visual Studio -> carpeta -> MIA_Semana14 -> git clone [repo url] -> dotnet commands

NOTA: cambia ren .csproj file -> <TargetFramework>net9.0</TargetFramework>



```
samanthaarrazola@Samanthas-MacBook-Air ~ % cd ~/Documents
samanthaarrazola@Samanthas-MacBook-Air Documents % mkdir MIA_Semana14
samanthaarrazola@Samanthas-MacBook-Air Documents % cd MIA_Semana14
samanthaarrazola@Samanthas-MacBook-Air MIA_Semana14 % git clone https://github.com/lsarrazola/serviceconnector-webapp-storageblob-dotnet.git
Cloning into 'serviceconnector-webapp-storageblob-dotnet'...
remote: Enumerating objects: 124, done.
remote: Counting objects: 100% (124/124), done.
remote: Compressing objects: 100% (102/102), done.
remote: Total 124 (delta 40), reused 77 (delta 16), pack-reused 0 (from 0)
Receiving objects: 100% (124/124), 700.28 KiB | 1.94 MiB/s, done.
Resolving deltas: 100% (40/40), done.
samanthaarrazola@Samanthas-MacBook-Air MIA_Semana14 % ls
serviceconnector-webapp-storageblob-dotnet
samanthaarrazola@Samanthas-MacBook-Air MIA_Semana14 % cd serviceconnector-webapp-storageblob-dotnet
samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % ls
CHANGELOG.md          Program.cs             WebStorageSample.csproj
CONTRIBUTING.md       Properties             appsettings.Development.json
Const.cs              README.md             appsettings.json
LICENSE.md             Startup.cs            wwwroot
Pages                 StorageHelper.cs
samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet %
```



3. Crear DotNet Env

a) En terminal, instalar: dotnet add package DotNetEnv

```
samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % dotnet add package DotNetEnv

Build succeeded in 0.3s
info : X.509 certificate chain validation will use the fallback certificate bundle at '/usr/local/share/dotnet/sdk/9.0.200/trustedroots/cod
esignctl.pem'.
info : X.509 certificate chain validation will use the fallback certificate bundle at '/usr/local/share/dotnet/sdk/9.0.200/trustedroots/tim
estampctl.pem'.
info : Adding PackageReference for package 'DotNetEnv' into project '/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp
-storageblob-dotnet/WebStorageSample.csproj'.
info : GET https://api.nuget.org/v3/registration5-gz-semver2/dotnetenv/index.json
info : OK https://api.nuget.org/v3/registration5-gz-semver2/dotnetenv/index.json 438ms
info : Restoring packages for /Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.cs
proj...
info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.10.06.23.47.38/vulnerability.base.json
info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.10.06.23.47.38/2026.10.07.11.47.40/vulnerability.update.json
warn : NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
warn : NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj
9
warn : NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wvxc-855f-jvr
v
warn : NU1902: Package 'Azure.Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9
hc6-r2h6
info : Package 'DotNetEnv' is compatible with all the specified frameworks in project '/Users/samanthaarrazola/Documents/MIA_Semana14/servi
ceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj'.
info : PackageReference for package 'DotNetEnv' version '3.2.0' added to file '/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnec
tor-webapp-storageblob-dotnet/WebStorageSample.csproj'.
info : Writing assets file to disk. Path: /Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/obj/pro
ject.assets.json
log : Restored /Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj (in 255 m
s).
samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet %
```



b) Crear .env file en raíz de proyecto, valores ->

AZURE_STORAGEBLOB_RESOURCEENDPOINT=https://miarchivos2026.blob.core.windows.net/
AZURE_STORAGE_CONTAINER_NAME=default

The screenshot shows the Visual Studio Code interface. The Explorer pane on the left shows the project structure with files like .env, .env.production, .gitignore, appsettings.json, CHANGELOG.md, Const.cs, CONTRIBUTING.md, LICENSE.md, Program.cs, README.md, Startup.cs, and StorageHelper.cs. The .env file is selected. The main editor shows the content of .env:

```
1 AZURE_STORAGEBLOB_RESOURCEENDPOINT=https://miarchivos2026.blob.core.windows.net/  
2 AZURE_STORAGE_CONTAINER_NAME=default
```

The Output pane at the bottom shows the terminal output of the command `dotnet add package DotNetEnv`. The output indicates that the package was successfully added and restored, and it lists the versions of the packages and their dependencies.

c) Crear .env.production con los mismos valores en el mismo lugar

The screenshot shows the Visual Studio Code interface. The Explorer pane on the left shows the project structure with files like .env, .env.production, .gitignore, appsettings.json, CHANGELOG.md, Const.cs, CONTRIBUTING.md, LICENSE.md, Program.cs, README.md, Startup.cs, and StorageHelper.cs. The .env.production file is selected. The main editor shows the content of .env.production:

```
1 AZURE_STORAGEBLOB_RESOURCEENDPOINT=https://miarchivos2026.blob.core.windows.net/  
2 AZURE_STORAGE_CONTAINER_NAME=default
```

The Output pane at the bottom shows the terminal output of the command `dotnet add package DotNetEnv`. The output indicates that the package was successfully added and restored, and it lists the versions of the packages and their dependencies.



d) Anadir cambio brindado en program.cs

```

1 | using DotNetEnv;
2 | using Microsoft.AspNetCore.Hosting;
3 | using Microsoft.Extensions.Hosting;
4 |
5 | namespace WebStorageSample
6 | {
7 |     0 references
8 |     public class Program
9 |     {
10 |         0 references
11 |         public static void Main(string[] args)
12 |         {
13 |             Env.Load("*.env");
14 |             CreateHostBuilder(args).Build().Run();
15 |         }
16 |
17 |         1 reference
18 |         public static IHostBuilder CreateHostBuilder(string[] args) =>
19 |         {
20 |             Host.CreateDefaultBuilder(args)
21 |                 .ConfigureWebHostDefaults(webBuilder =>
22 |                 {
23 |                     webBuilder.UseStartup<Startup>();
24 |                 })
25 |                 .Build();
26 |         }
27 |     }
28 | }

```

e) dotnet clean -> dotnet restore -> dotnet build -> dotnet run

```

samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % dotnet clean
Build succeeded in 0.2s
samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % dotnet restore
Restore succeeded with 4 warning(s) in 0.2s
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj9
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wvxc-855f-jrvv
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
Build succeeded with 4 warning(s) in 0.3s
samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % dotnet build
Restore succeeded with 4 warning(s) in 0.2s
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj9
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wvxc-855f-jrvv
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
WebStorageSample succeeded with 6 warning(s) (2.5s) -> bin/Debug/net9.0/WebStorageSample.dll
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj9
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wvxc-855f-jrvv
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/StorageHelper.cs(34,17): warning CA2200: Re-throwing caught exception changes stack information (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2200)
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/StorageHelper.cs(66,17): warning CA2200: Re-throwing caught exception changes stack information (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2200)
Build succeeded with 10 warning(s) in 2.9s
samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % dotnet run
Using launch settings from /Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/Properties/launchSettings.json...
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj9
/Users/samanthaarrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package

```



```
Successfully created and trusted a new HTTPS certificate.
samantharrazola@Samanthas-MacBook-Air: serviceconnector-webapp-storageblob-dotnet % dotnet run
Using launch settings from /Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/Properties/launchSettings.json...
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
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/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj9
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wvxc-855f-jvrv
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: https://localhost:5001
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5000
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: /Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet
```

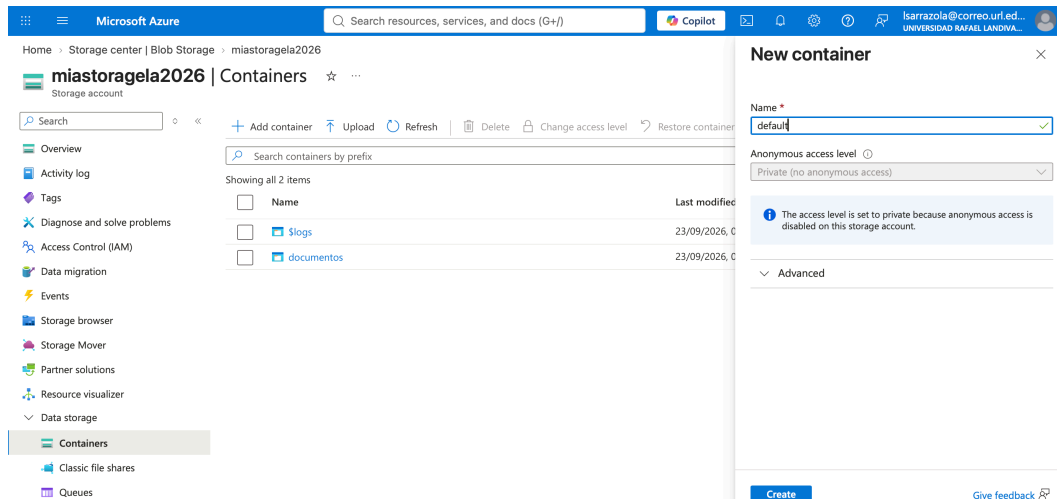
4. Crear recursos en Azure

En azure portal site, cree o utilice un Resource Group, una Storage Account con Blob Storage y un contenedor. El contenedor deberá tener al menos tres archivos de prueba, Puede utilizar el creado en semana 13.

5. Conectar App Service con Blob Storage

Configure la conexión entre App Service y Blob Storage mediante Service Connector utilizando identidad administrada. Verifique que la identidad tenga los permisos necesarios sobre los blobs.

- Azure Portal → **Storage accounts** → **tu Storage Account** → **Data storage** → **Containers**
- Crear contenedor “default”





- c) Activa identidad administrada en App Service: **Azure Portal** → **App Service** → **Settings** → **Identity** → **System assigned: on** → **Save** (web app creada semana 13)

Microsoft Azure Portal - Create Web App

Home > App Services

Basics Database Deployment Networking Monitor + secure Tags Review + create

App Service Web Apps lets you quickly build, deploy, and scale enterprise-grade web, mobile, and API apps running on any platform. Meet rigorous performance, scalability, security and compliance requirements while using a fully managed platform to perform infrastructure maintenance. [Learn more](#)

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource Group * rg-mia-lab [Create new](#)

Instance Details

Name storage-container-connector
-gagza3gph9d9gdj.northcentralus-01.azurewebsites.net

Publish * ☒ Code ☐ Container

Runtime stack * .NET 9 (STS)
⚠ .NET 9 (STS) will reach EOL on 10/11/2026 [Learn More](#)

Operating System * ☐ Linux ☒ Windows

Region * North Central US

[Review + create](#) < Previous Next: Database >

Microsoft Azure Portal - Microsoft.Web-WebApp-Portal-d572114e-98ab | Overview

Home

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

Your deployment is complete

Deployment name : Microsoft.Web-WebApp-Portal-d572114e-98ab
Subscription : Azure for Students
Resource group : rg-mia-lab
Start time : 07/10/2026, 08:11:23
Correlation ID : cde72a4c-98ad-4676-8d24-00b10c43c3af

> Deployment details

✓ Next steps

[Go to resource](#)

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)



- d) Configurar la conexión entre App Service y Blob Storage: **App Service** → **Settings** → **Service Connector** → **Create** → **Seleccionar “Storage Blob”** → **Seleccionar Storage Account [tu_blob_name]** → **Authentication “System-assigned managed identity”** → **Review + Create**.



Microsoft Azure Portal - storage-container-connector | Service Connector

Search resources, services, and docs (Google)

Home > storage-container-connector | Service Connector

Using Service Connector will register resource provider 'Microsoft.ServiceLinker' to your Subscription.

Authentication

System assigned managed identity

Next: Networking

Microsoft Azure Portal - storage-container-connector | Service Connector

Search resources, services, and docs (Google)

Home > storage-container-connector | Service Connector

Using Service Connector will register resource provider 'Microsoft.ServiceLinker' to your Subscription.

Service type: Storage - Blob

Resource name: miastorage2026

Connection name: storageblob_1sae

Connection status: Unvalidated

Microsoft Azure Portal - storage-container-connector | Service Connector

Search resources, services, and docs (Google)

Home > storage-container-connector | Service Connector

Using Service Connector will register resource provider 'Microsoft.ServiceLinker' to your Subscription.

Service type: Storage - Blob

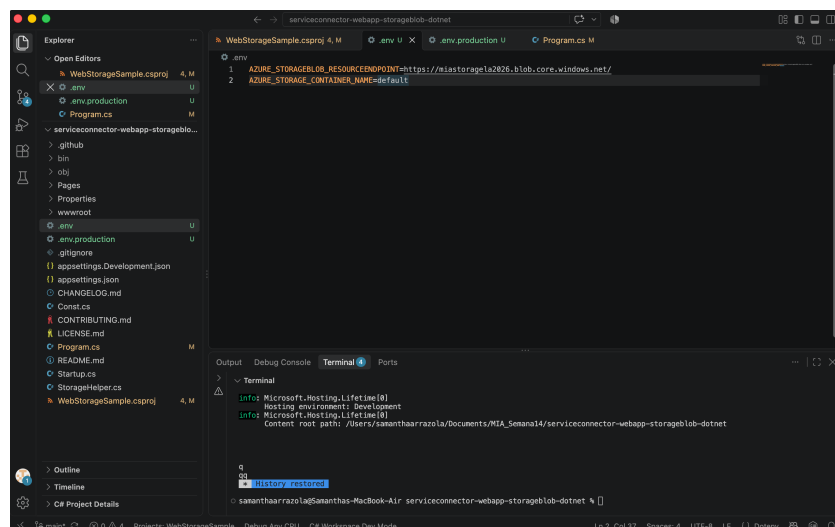
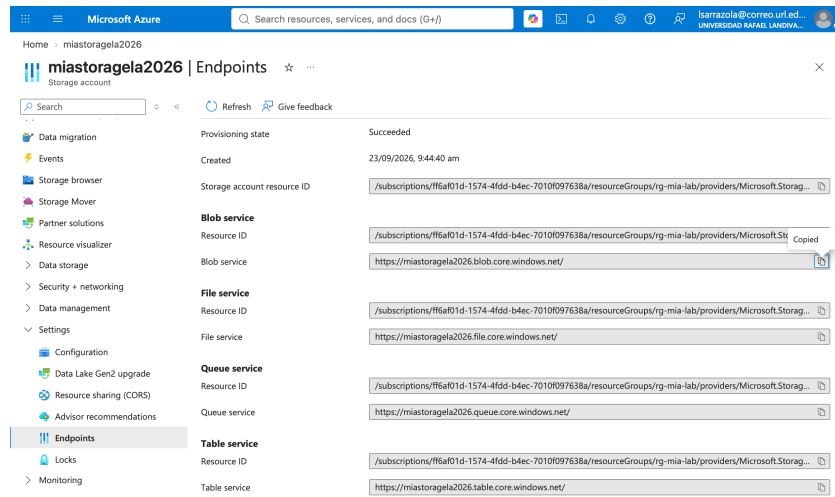
Resource name: miastorage2026

Connection name: storageblob_1sae

Connection status: Success



- e) Reemplazar storage blob endpoint y container nombre en .env file
Storage Account → Settings → Endpoints → Blob service



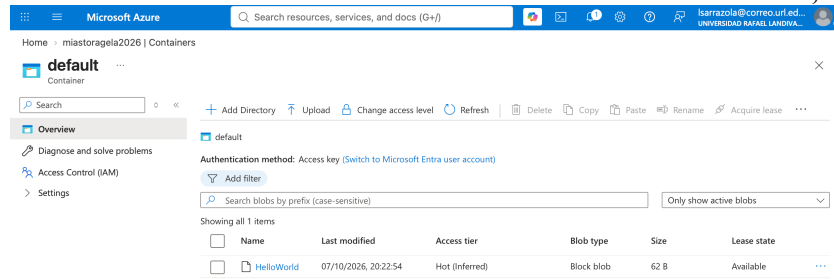
6. Probar el acceso al almacenamiento:

Ejecute la aplicación local y compruebe que puede leer los blobs. Realice al menos una operación de carga de archivo y verifique desde Azure Portal que el archivo llegó al contenedor. Visual Studio -> dotnet run

WebStorageSample Home Privacy

Hello Service Connector! UTC Now: 8/10/2026
2:22:51?AM +00:00.

Learn about building Web apps with ASP.NET Core.



- Verifique que Azure CLI esté instalado ejecutando en la terminal de VS Code: `az --version`
- Inicie sesión en Azure: `az login`

```

samantha@razzola:~$ samantha-MacBook-Air: serviceconnector-webapp-storageblob-dotnet % az --version
azure-cli                        2.9.0.0 *

core                             2.90.0 *
telemetry                       1.1.0

Dependencies:
msal                            1.36.0
azure-rgctl-resouce             24.0.0

Python location /opt/homebrew/Cellar/azure-cli/2.90.0/libexec/bin/python
Config directory /Users/samantha@razzola/.azure
Extensions directory /Users/samantha@razzola/.azure/cliextensions/

Python (Darwin) 3.14.8 (main, Sep 30 2026, 17:55:09) [Clang 12.0.0 (clang-1200.3.34.2)]

Legal docs and information: aka.ms/AzureCLILegal

You have 2 updates(s) available. Consider updating your CLI installation with "az upgrade"
samantha@razzola:~$ samantha-MacBook-Air: serviceconnector-webapp-storageblob-dotnet % az login
A web browser has been opened at https://login.microsoftonline.com/organizations/0a27b7v2.0/authorize. Please continue the login in the
b browser. If no web browser is available or if the web browser fails to open, use device code flow with az login --use-device-code

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No | Subscription name | Subscription ID | Tenant
---|---|---|---
[1] * Azure for Students | ffe610d-1574-4fdd-b4ec-7010f09763ba | Universidad Rafael Landívar

The default is marked with an (*); the default tenant is "Universidad Rafael Landívar" and subscription is "Azure for Students" (ffe610d-1574-b4ec-7010f09763ba)

Select a subscription and tenant (Type a number or Enter for no changes): 1

Tenant: Universidad Rafael Landívar
Subscription: Azure for Students (ffe610d-1574-4fdd-b4ec-7010f09763ba)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configura
tion at https://go.microsoft.com/fwlink/?linkid=2277238

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

> samantha@razzola:~$ samantha-MacBook-Air: serviceconnector-webapp-storageblob-dotnet %

```

- c) Verifique la cuenta de Azure activa: az account show

```

samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % az account show
{
  "environmentName": "AzureCloud",
  "homeTenantId": "0f78549d-3eec-43af-b56a-6f7b042d6c9a",
  "id": "fff6af01d-1574-4fdd-b4ec-7010f097638a",
  "isDefault": true,
  "managedByTenants": [],
  "name": "Azure for Students",
  "state": "Enabled",
  "tenantDefaultDomain": "correo.url.edu.gt",
  "tenantDisplayName": "Universidad Rafael Landivar",
  "tenantId": "0f78549d-3eec-43af-b56a-6f7b042d6c9a",
  "user": {
    "name": "lsarrazola@correo.url.edu.gt",
    "type": "user"
  }
}

samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet %

```

- d) Verifique que pueda obtener un token para Azure Storage: `az account get-access-token --resource https://storage.azure.com/`

[illegible]



e) Si se muestra un accessToken la autenticación esta correcta, ejecutar -> dotnet run

```
samantharrazola@Samanthas-MacBook-Air: serviceconnector-webapp-storageblob-dotnet % dotnet run
Using launch settings from /Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/Properties/launchSettings.json...
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5v-6r4h-3vj9
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wxwc-855f-jrvv
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5v-6r4h-3vj9
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wxwc-855f-jrvv
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: https://localhost:5001
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5000
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: /Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet
Hello Service Connector! UTC Now: 8/10/2026 2:30:127AM +00:00.
```

f) Si hay error en acceso, ejecutar en terminal vs code: az role assignment create --assignee "TU_CORREO_DE_AZURE" --role "Storage Blob Data Contributor" --scope "/subscriptions/TU_SUBSCRIPTION_ID/resourceGroups/TU_RESOURCE_GROUP/providers/Microsoft.Storage/storageAccounts/TU_STORAGE"

Obtener info: az account show --query "{subscriptionId:id, user:user.name}" -o table
az storage account show --name "tu_storage" --query "{id:id, resourceGroup:resourceGroup}" -o table

g) Cerrar process ctr +c y logear de nuevo: az logout y az login, Por ultimo dotnet run

```
samantharrazola@Samanthas-MacBook-Air: serviceconnector-webapp-storageblob-dotnet % az login
[!] * Azure for Students f66af01d-1574-4fdd-b4ec-7010f097638a Universidad Rafael Landivar
The default is marked with an *: the default tenant is 'Universidad Rafael Landivar' and subscription is 'Azure for Students' (f66af01d-1574-4fdd-b4ec-7010f097638a).

Select a subscription and tenant (Type a number or Enter for no changes): 1
Tenant: Universidad Rafael Landivar
Subscription: Azure for Students (f66af01d-1574-4fdd-b4ec-7010f097638a)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

samantharrazola@Samanthas-MacBook-Air: serviceconnector-webapp-storageblob-dotnet % dotnet run
Using launch settings from /Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/Properties/launchSettings.json...
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5v-6r4h-3vj9
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wxwc-855f-jrvv
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wcx-rv27
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5v-6r4h-3vj9
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wxwc-855f-jrvv
/Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: https://localhost:5001
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5000
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: /Users/samantharrazola/Documents/MIA_Semana14/serviceconnector-webapp-storageblob-dotnet
Hello Service Connector! UTC Now: 8/10/2026 2:37:187AM +00:00.
```

Nota: no olvidar subir cambios a repo github fork



```
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   Program.cs
    modified:   WebStorageSample.csproj

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .env
    .env.production

no changes added to commit (use "git add" and/or "git commit -a")
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git add .
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   .env
    new file:   .env.production
    modified:   Program.cs
    modified:   WebStorageSample.csproj

● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git commit "feat(blob): añadiendo .env file para autent
lación de blob"
error: pathspec 'feat(blob): añadiendo .env file para autenticación de blob' did not match any file(s) known to git
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git commit -m "feat(blob): añadiendo .env file para autenticación de blob"
[main 084307] feat(blob): añadiendo .env file para autenticación de blob
 4 files changed, 12 insertions(+), 10 deletions(-)
 create mode 100644 .env
 create mode 100644 .env.production
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 662 bytes | 662.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/Lsarrazola/serviceconnector-webapp-storageblob-dotnet.git
 4be51c7..084307 main -> main
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet %
```

 .env	feat(blob): añadiendo .env file para autenticación de bl...	1 minute ago
 .env.production	feat(blob): añadiendo .env file para autenticación de bl...	1 minute ago
 .gitignore	initial version	5 years ago
 CHANGELOG.md	CHANGELOG.md committed	5 years ago
 CONTRIBUTING.md	CONTRIBUTING.md committed	5 years ago
 Const.cs	initial version	5 years ago
 LICENSE.md	LICENSE.md committed	5 years ago
 Program.cs	feat(blob): añadiendo .env file para autenticación de bl...	1 minute ago

Explorer

WebStorageSample.csproj 4 · .gitignore · .env · .env.production · Program.cs

Open Editors

WebStorageSample.csproj 4

.gitignore

.env

.env.production

Program.cs

serviceconnector-w...

...

Pages

Properties

wwwroot

.env

.env.production

.gitignore

appsettings.Develo...

appsettings.json

CHANGELOG.md

Const.cs

CONTRIBUTING.md

LICENSE.md

Program.cs

README.md

Startup.cs

StorageHelpers.cs

WebStorageSa...

Outline

No symbols found in document 'gitignore'

Timeline

C# Project Details

Output

Debug Console

Terminal

Ports

Terminal

● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git status

On branch main

Your branch is up to date with 'origin/main'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: .gitignore

no changes added to commit (use "git add" and/or "git commit -a")

● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git add .gitignore

● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git status

On branch main

Your branch is up to date with 'origin/main'.

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

modified: .gitignore

● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git commit -m "feat(blob): añadiendo .env al .gitignore"

[main 8f4b74] feat(blob): añadiendo .env al .gitignore

1 file changed, 2 insertions(+), 1 deletion(-)

● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git push

Enumerating objects: 5, done.

Counting objects: 100% (5/5), done.

Delta compression using up to 8 threads

Compressing objects: 100% (3/3), done.

Writing objects: 100% (3/3), 326 bytes | 326.00 KiB/s, done.

Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)

remote: Resolving deltas: 100% (2/2), completed with 2 local objects.

To https://github.com/Lsarrazola/serviceconnector-webapp-storageblob-dotnet.git

084307..8f4b74 main -> main

● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet %



```

WebStorageSample.csproj 4
  .env
  .gitignore
  .env.production
  Program.cs

serviceconnector-w...
  .env
  .gitignore
  .env.production
  Program.cs

CHANGELOG.md
CONTRIBUTING.md
LICENSE.md
Program.cs
README.md
StorageHelpers.cs
WebStorageSample.csproj
WebStorageSa... 4
  Outline
  No symbols found
  In document '.env'

Timeline
  CS Project Details

Terminal
  saantharrazola@Santhas-MacBook-Air: serviceconnector-webapp-storageblob-dotnet % git status
  Changes not staged for commit:
    (use "git add <file>..." to update what will be committed)
    (use "git restore <file>..." to discard changes in working directory)
    modified:   .env

  no changes added to commit (use "git add" and/or "git commit -a")
  saantharrazola@Santhas-MacBook-Air: serviceconnector-webapp-storageblob-dotnet % dotnet build
  Restore succeeded with 4 warning(s) in 8.3s
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1
  h severity vulnerability, https://github.com/advisories/GSA-Safe-4ucx-v27
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1
  erate severity vulnerability, https://github.com/advisories/GSA-45vy-6r4b-3vj9
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1
  erate severity vulnerability, https://github.com/advisories/GSA-wxc-835f-jrvv
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1
  own moderate severity vulnerability, https://github.com/advisories/GSA-544x-9hcs-2h6
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1
  WebStorageSample succeeded with 6 warning(s) (3.7s) = 110.0mb/output
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.dll
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.Identity' 1.4.1
  h severity vulnerability, https://github.com/advisories/GSA-Safe-4ucx-v27
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1
  erate severity vulnerability, https://github.com/advisories/GSA-45vy-6r4b-3vj9
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1
  erate severity vulnerability, https://github.com/advisories/GSA-wxc-835f-jrvv
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.Identity' 1.4.1
  own moderate severity vulnerability, https://github.com/advisories/GSA-544x-9hcs-2h6
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/StorageHelper.cs(34,37): warning CA2294: Re-throwing caught exception cha
  rection. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2294)
  /Users/santharrazola/Documents/RIA_Semana14/serviceconnector-webapp-storageblob-dotnet/StorageHelper.cs(66,37): warning CA2294: Re-throwing caught exception cha
  rection. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2294)
  Build succeeded with 18 warning(s) in 4.5s
  saantharrazola@Santhas-MacBook-Air: serviceconnector-webapp-storageblob-dotnet %
  
```

7. Crear Pipeline Yml

En la raíz de tu fork de GitHub crear archivo “azure-pipelines.yml”

Colocar en el archivo lo siguiente:

trigger:

- main

pool:

vmImage: 'windows-latest'

variables:

buildConfiguration: 'Release'

dotnetVersion: '9.x'

Completar posteriormente

azureSubscription: 'NOMBRE-DE-TU-SERVICE-CONNECTION'

appName: 'NOMBRE-DE-TU-APP-SERVICE'

steps:

1. Instalar .NET 9

- task: UseDotNet@2

displayName: 'Instalar .NET 9 SDK'

inputs:

packageType: 'sdk'

version: '\$(dotnetVersion)'

2. Restaurar dependencias

- task: DotNetCoreCLI@2

displayName: 'Restaurar dependencias'

inputs:

command: 'restore'

projects: '**/*.csproj'

3. Configurar ambiente de producción

- powershell: |

Copy-Item ".env.production" ".env" -Force

displayName: 'Configurar .env de producción'



4. Compilar

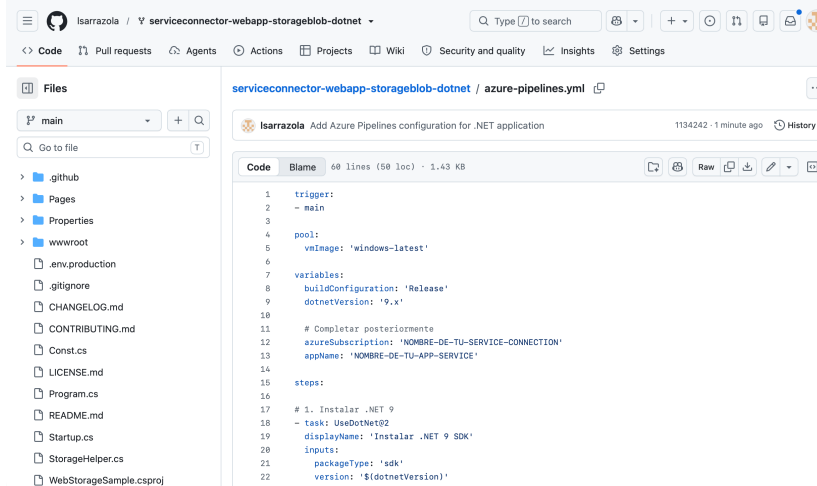
```
- task: DotNetCoreCLI@2
  displayName: 'Compilar aplicación'
  inputs:
    command: 'build'
    projects: '**/*.csproj'
    arguments: '--configuration $(buildConfiguration) --no-restore'
```

5. Publicar

```
- task: DotNetCoreCLI@2
  displayName: 'Publicar aplicación'
  inputs:
    command: 'publish'
    projects: '**/*.csproj'
    arguments: '--configuration $(buildConfiguration) --output $(Build.ArtifactStagingDirectory)/publish --no-build'
    zipAfterPublish: true
```

6. Desplegar en Azure App Service

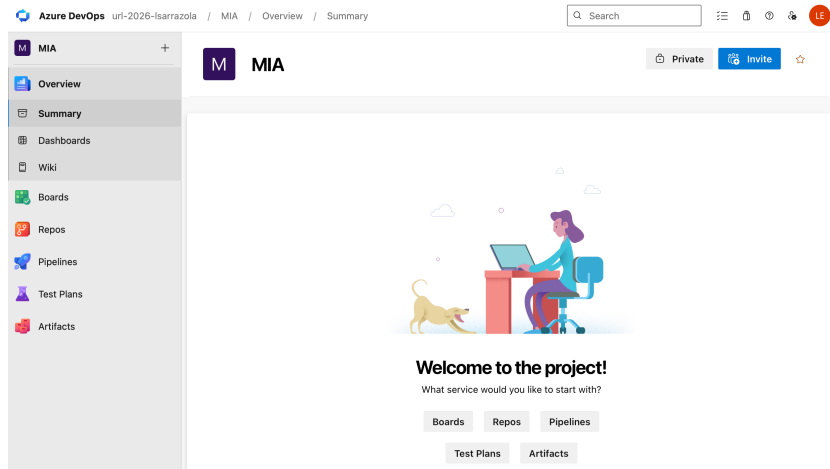
```
- task: AzureWebApp@1
  displayName: 'Desplegar en Azure App Service'
  inputs:
    azureSubscription: '$(azureSubscription)'
    appType: 'webApp'
    appName: '$(appName)'
    package: '$(Build.ArtifactStagingDirectory)/publish/**/*.zip'
```



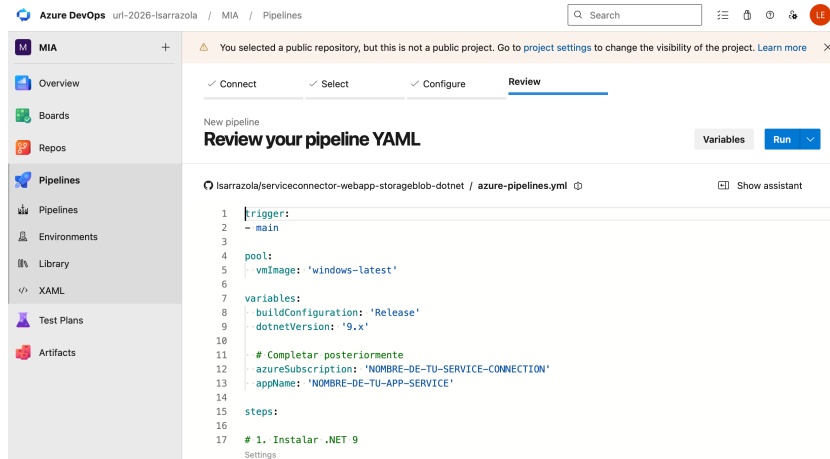
8. Crear pipeline CI/CD

Cree un pipeline YAML que restaure dependencias, compile, publique y despliegue la aplicación en Azure App Service.

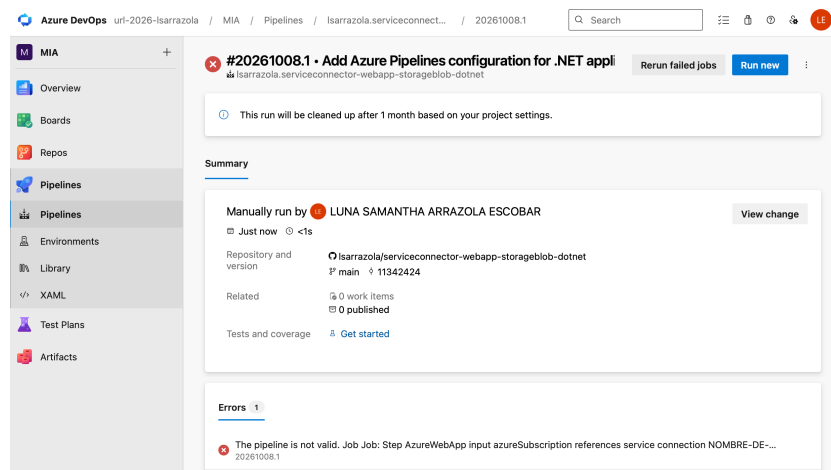
- a) Conectarse a Azure DevOps con misma cuenta que azure portal -> <https://aex.dev.azure.com/me>



- b) Crear pipeline: En Azure devOps -> Menu izquierda -> pipelines -> Create Pipeline -> Seleccionar Github -> Autoriza Azure DevOps para acceder a GitHub -> seleccionar repositorio que hizo fork -> seleccionar Existing Azure Pipelines YAML file -> Branch: main Path: /azure-pipelines.yml

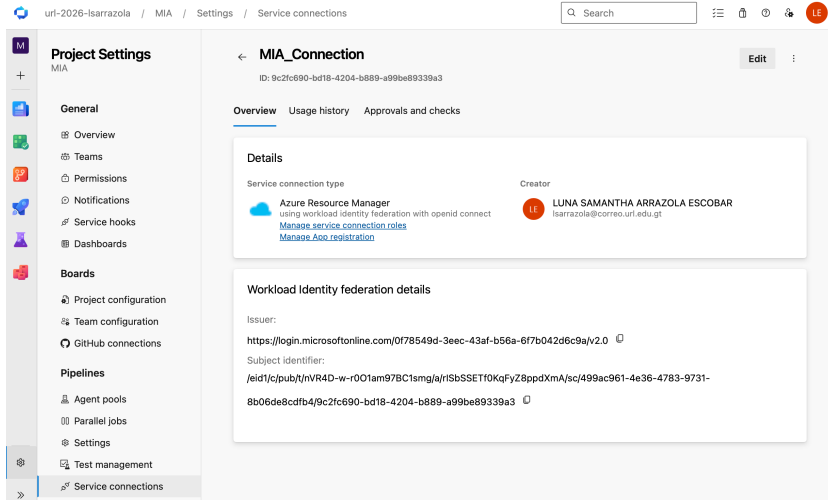


- c) Probar seleccionando “run”



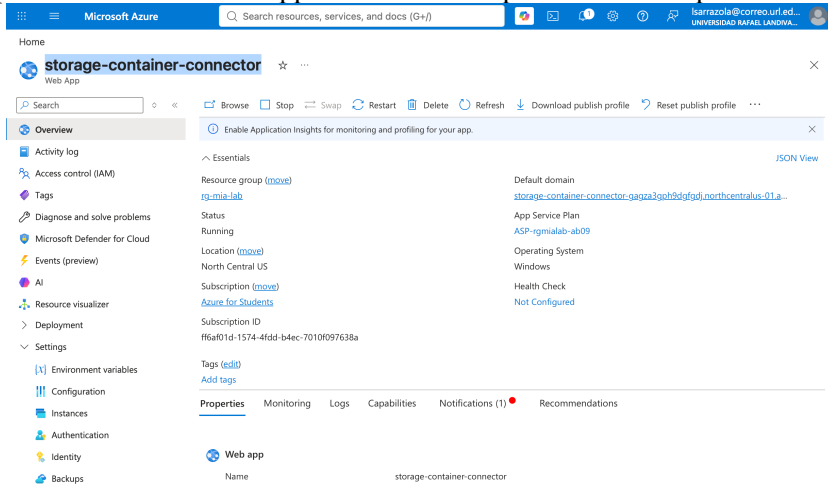


- d) Conectar Azure DevOps con Azure Portal: **Project Settings** → **Service connections** → **New service connection** -> **Azure Resource Manager** -> luego, **Workload identity federation** -> **Selecciona tu suscripción de Azure** -> y **resource group** donde estan tus recursos -> **Anadir nombre “MIA_Connection”** -> Seleccionar **“grant access permission to all pipelines”** -> **Save**



9. Conectar Pipeline con App

- a) Obtener nombre app: En Azure Portal → App Services → tu aplicación → Copia nombre app



- b) En github en tu “azure-pipelines.yml” file busca: azureSubscription y appName, cambiar por valores reales.

azureSubscription: TU_Service_Connection_name
appName: Tu_app_name



The image shows a file explorer on the left with a tree view containing folders like 'github', 'Pages', 'Properties', and 'wwwroot', and files like '.env.production', '.gitignore', 'CHANGELOG.md', 'CONTRIBUTING.md', 'Const.cs', 'LICENSE.md', 'Program.cs', 'README.md', 'Startup.cs', 'StorageHelper.cs', 'WebStorageSample.csproj', 'appsettings.Development.json', 'appsettings.json', and 'azure-pipelines.yml'. The 'azure-pipelines.yml' file is selected. On the right, the code editor displays the content of 'azure-pipelines.yml' for the 'serviceconnector-webapp-storageblob-dotnet' project. The file is 60 lines long, 50 loc, and 1.41 KB. It defines a pipeline with a trigger for 'main', a pool with 'windows-latest' VM image, and two steps: '1. Instalador .NET 9' and '2. Restaurar dependencias'.

c) Al hacer commit, verificar el pipeline se ejecuta automáticamente

```
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git pull origin
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 2.48 KiB | 317.00 KiB/s, done.
From https://github.com/lsarrazola/serviceconnector-webapp-storageblob-dotnet
  448c60f..91b2835  main    -> origin/main
Updating 448c60f..91b2835
Fast-forward
 azure-pipelines.yml | 60 ++++++
 1 file changed, 60 insertions(+)
 create mode 100644 azure-pipelines.yml
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   azure-pipelines.yml

no changes added to commit (use "git add" and/or "git commit -a")
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git add azure-pipelines.yml
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git commit -m "feat(blob): pipeline yaml change"
1 file changed, 1 insertion(+), 1 deletion(-)
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 321 bytes | 321.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/lsarrazola/serviceconnector-webapp-storageblob-dotnet.git
  91b2835..9f8b144  main -> main
○ samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet %
```

The image shows a GitHub repository page for 'lsarrazola / serviceconnector-webapp-storageblob-dotnet'. The commit 9f8b144 is highlighted, showing it was committed 24 minutes ago. The commit message is 'feat(blob): pipeline yaml change'. Below the commit, a diff view shows the changes in 'azure-pipelines.yml'. The diff highlights the addition of a new step, '2. Restaurar dependencias', and the removal of a comment. The diff shows the following changes:

```
11 - # Completar posteriormente
11 + # Completar posteriormente
12   azureSubscription: 'MIA_Connection'
13   appName: 'storage-container-connector'
```



10. Validar la solución

Compruebe desde la URL pública que la aplicación funciona y que el acceso al Blob Storage continúa operativo después del despliegue.

The screenshot shows the Azure DevOps interface for a pipeline named 'Desplegar en Azure App Service'. The pipeline is successful, and the output shows the deployment details for the 'WebStorageSample' application. The output includes the evaluation of the deployment, the selection of the Node version, and the deployment of the application to the Azure App Service. The output also shows the application settings and the deployment URL.

Hello Service Connector! UTC Now: 10/8/2026
4:09:03 AM +00:00.

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- acceso de la aplicación al Blob: Azure Portal → App Service → Settings → Environment variables → App settings
-> colocar `AZURE_STORAGEBLOB_RESOURCEENDPOINT`
https://tu_storage.blob.core.windows.net/
(si existe obviar)
- Verificar app: Azure Portal → App Service → OverView → Default Domain → click en URL publica
- En la app publicada realizar cambios en archivos
- Verificar que los cambios se representen en storage account: Azure Portal → Storage Account → Storage Browser
→ Blob containers → Seleccionar su container → Verificar cambios



```
samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % dotnet build
/Users/samanthaarrazola/Documents/MIA_Semanal4/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.
Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
WebStorageSample succeeded with 6 warning(s) (0.7s) - bin/Debug/net9.0/WebStorageSample.dll
/Users/samanthaarrazola/Documents/MIA_Semanal4/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1903: Package 'Azure.
Identity' 1.4.1 has a known high severity vulnerability, https://github.com/advisories/GHSA-5mfx-4wxc-rv27
/Users/samanthaarrazola/Documents/MIA_Semanal4/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.
Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-5dyv-6rhh-3vj9
/Users/samanthaarrazola/Documents/MIA_Semanal4/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.
Identity' 1.4.1 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wxc-855f-jrvy
/Users/samanthaarrazola/Documents/MIA_Semanal4/serviceconnector-webapp-storageblob-dotnet/WebStorageSample.csproj : warning NU1902: Package 'Azure.
Storage.Blobs' 12.10.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
/Users/samanthaarrazola/Documents/MIA_Semanal4/serviceconnector-webapp-storageblob-dotnet/StorageHelper.cs(34,17): warning CA2200: Re-throwing caught
exception changes stack information (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2200)
/Users/samanthaarrazola/Documents/MIA_Semanal4/serviceconnector-webapp-storageblob-dotnet/StorageHelper.cs(66,17): warning CA2200: Re-throwing caught
exception changes stack information (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2200)

Build succeeded with 10 warning(s) in 1.2s
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Pages/Index.cshtml
        modified:   Pages/Index.cshtml.cs

no changes added to commit (use "git add" and/or "git commit -a")
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git add .
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git commit -m "feat(blob): añadiendo funcionalidad carga de imágenes"
[main a6593bd] feat(blob): añadiendo funcionalidad carga de imágenes
2 files changed, 92 insertions(+), 17 deletions(-)
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet % git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 1.63 KiB | 1.63 MiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/lsarrazola/serviceconnector-webapp-storageblob-dotnet.git
9f8b144..a6593bd main -> main
● samanthaarrazola@Samanthas-MacBook-Air serviceconnector-webapp-storageblob-dotnet %
```

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Description	Stages	
✓ #20261008.3 - feat(blob): añadiendo funcionalidad carga d...	✓	8m ago 2m 26s
Individual CI for main a6593bd		
✓ #20261008.2 - feat(blob): pipeline yaml change	✓	1h ago 3m 26s
Manually run by main 9f8b144		

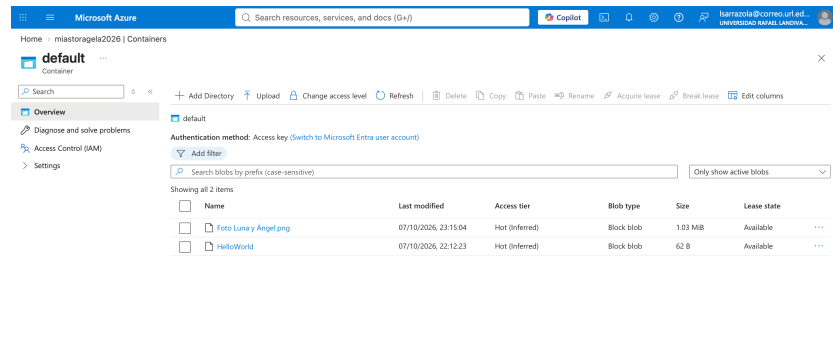
WebStorageSample Home Privacy

Azure Blob Storage

Seleccione una imagen para almacenarla en Azure Blob Storage.

Choose File no file selected

Subir imagen



11. Preguntas de analisis

1. ¿Para qué sirve el archivo .env en la aplicación?

El archivo .env sirve para guardar las variables de configuración que necesita la aplicación para funcionar. En nuestro caso lo usamos para indicar la dirección de Azure Blob Storage y el nombre del contenedor donde se almacenan los archivos. Esto permite cambiar la configuración sin tener que modificar directamente el código.

2. ¿Por qué utilizar Blob Storage y no guardar los archivos dentro del proyecto?

Porque Blob Storage permite guardar los archivos en la nube de forma más segura y organizada. Asimismo, los archivos no se pierden si actualizamos o volvemos a publicar la aplicación. También podemos acceder a ellos sin depender del almacenamiento del servidor donde está funcionando el proyecto.

3. ¿Cuál es la diferencia entre el endpoint de Blob Storage y la URL de App Service?

El endpoint de Blob Storage es la dirección que utiliza la aplicación para acceder a los archivos almacenados en Azure. En cambio, la URL de App Service es la dirección que utilizamos para abrir nuestra aplicación desde un navegador. Básicamente, una sirve para acceder al almacenamiento y la otra para acceder a la página web.

4. ¿Por qué utilizar una identidad administrada en lugar de una Storage Account Key?

Porque la identidad administrada permite que la aplicación se conecte a Azure Storage sin necesidad de guardar claves o contraseñas en el código. Esto hace que la conexión sea más segura y facilita la administración de los permisos, ya que Azure se encarga de manejar la autenticación.

5. ¿Qué función cumplen GitHub, Azure DevOps y la Service Connection?

GitHub sirve para guardar nuestro código y llevar un control de los cambios que realizamos. Azure DevOps se encarga de ejecutar el pipeline para compilar y publicar automáticamente la aplicación. Por último, la Service Connection permite que Azure DevOps se conecte de forma segura con nuestra suscripción de Azure para realizar el despliegue.

6. ¿Qué ocurre después de hacer push a main?

Cuando hacemos push a la rama main, los cambios se suben a GitHub y se activa automáticamente el pipeline de Azure DevOps, si está configurado correctamente. El pipeline descarga el código actualizado, restaura las dependencias, compila la aplicación y la publica en Azure App Service. De esta manera, no tenemos que realizar todo el proceso manualmente cada vez que hacemos un cambio.

7. Si el pipeline termina correctamente pero la aplicación recibe un 403 al subir un archivo, ¿qué revisarías?

Primero revisaría que la identidad administrada de App Service esté activada y que tenga los permisos necesarios para trabajar con Blob Storage, como el rol Storage Blob Data Contributor. También verificaría que el endpoint y el nombre del contenedor estén configurados correctamente. Esto se debe a que un pipeline puede terminar correctamente, pero la aplicación todavía puede tener problemas de permisos al intentar acceder al almacenamiento.



Entrega

Dentro del repositorio de GitHub, cree la carpeta semana-14/lab1/ y coloque en ella un PDF con la solución del laboratorio. Incluya también los archivos de respaldo generados durante la práctica, así como los pantallazos que evidencien las pruebas realizadas.

Enviar link de github a plataforma de la universidad.